

Background
Motivation

Abrupt Sunlight Reduction Scenarios (ASRS) represent low-probability but high-consequence global risks capable of causing severe disruptions to agricultural production. These scenarios may arise from large volcanic eruptions, nuclear conflict, or other events that inject light-blocking particles into the stratosphere. Although volcanic eruptions primarily generate sulfate aerosols and nuclear conflict produces black carbon soot, both mechanisms can reduce surface sunlight and crop yields. Argentina is one of the world's major food-exporting countries, making its capacity to maintain food production under ASRS conditions relevant not only for domestic food security but also for regional and global resilience. This study evaluates how resilient food strategies could mitigate food shortages under abrupt sunlight-reduction scenarios.

Methods
Methodology

Model framework
Adapted from the ALLFED Integrated Food System Model (Rivers et al., 2023), implemented in Python. Linear optimization minimizes starvation while maximizing caloric availability across the population, distributing food stocks and new production monthly over 120 months.
Crop yield loss estimates drawn from Xia et al. (2022)¹, covering six particulate loading levels (5–150 Tg). Baseline production data from FAO (2020–2022).
¹ Yield loss estimates are drawn from nuclear ASRS simulations (Xia et al., 2022). Volcanic-specific crop impact models for the Southern Hemisphere were not considered in this study. These figures are used as proxies and represent a key uncertainty in this analysis.

Research aim
Objective

To evaluate the effectiveness of a phased portfolio of adaptive food-system interventions on per-capita caloric availability in Argentina across a seven-year post-eruption window, modeled under ASRS conditions spanning 5 to 150 Tg¹
Specific aims:

1. Quantify production loss under volcanic ASRS without adaptation
2. Model phased interventions and their combined caloric yield
3. Assess regional food-security and export capacity implications

Scenarios modeled		
Scenario	Description	Adapt.
1a / 1b	Pre-event baseline (net / gross)	none
2a	150 Tg, no adaptation	none
2b	Redirection from livestock & biofuels	partial
2c	+ resilient food solutions	moderate
2d	+ greenhouse scale-up & area expansion	full

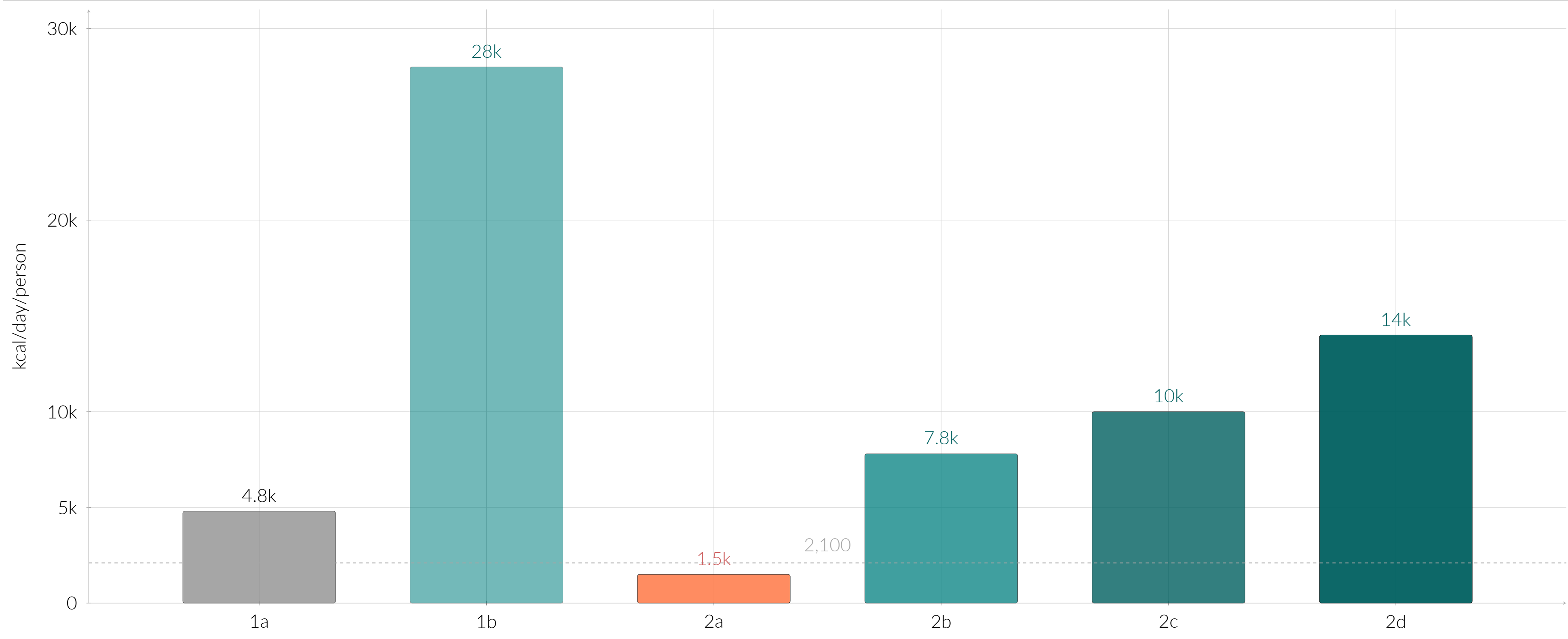
Country selection via multi-criteria decision matrix: supply-chain resilience (WEF GCI 2020), governance effectiveness (World Bank WGI 2021), and ASRS food production capacity.

Adaptive strategies
Interventions evaluated

Livestock & biofuel redirection Scaling down cattle and biofuel production releases grain currently consumed by animals. Estimated release: up to ~15,000 kcal/capita/day. Single highest-impact intervention (+300% availability vs. scenario 2a).	Cold-tolerant crop relocation Introducing northern-hemisphere cold-tolerant varieties (wheat, potato, oat, rapeseed) to latitudes where soy and maize yields collapse. Relocation should begin immediately after event.	Low-tech greenhouse scale-up Rapid expansion from 6,500 ha to ~2.3M ha within 2 years (×360). Controls temperature variability under reduced insolation. Scenario 2d key driver.	Cultivated area expansion A separate sensitivity analysis suggests that expansion up to 108 Mha (beyond the 72 Mha assumed in Scenario 2d) could theoretically sustain 6–22× Argentina's population.
Seaweed cultivation revival Argentine seaweed industry nearly extinct since mid-20th century. High potential along Chubut coast. Knowledge transfer from countries with active algae industries required.	Rationing & waste reduction Assumed in baseline model: equitable distribution across population; reduction of food loss due to scarcity. Necessary for all scenarios to prevent unequal access collapse.	Industrial sugar (lignocellulosic) Rapid repurposing of pulp and paper mills into lignocellulosic sugar factories. Contributes ~200 kcal/person/day in months 15–35 (scenario 2c/2d).	Single-cell protein (SCP) Methane-based SCP production to compensate animal protein losses. Used as animal feed substitute in scenarios 2c/2d to free crop production for human consumption.

■ Supply redirection & agricultural adaptation ■ Industrial & high-technology solutions

Results
Caloric availability by scenario



1a: Scenario 1a — Pre-event net production (~ 4,800 kcal/day/person). Represents actual food available for human consumption after accounting for livestock and biofuel use. Already 2× the minimum requirement.
1b: Scenario 1b — Pre-event gross production (~ 28,000 kcal/day/person). Includes all food produced including what goes to animals and biofuels. Illustrates the enormous potential for redirection in a crisis.
2a: Scenario 2a — 150 Tg ASRS, no adaptation (~ 1,500 kcal/day/person). Famine conditions: production falls below the 2,100 minimum. Only 71% of the population's caloric needs are met. Immediate action is required.
2b: Scenario 2b — Livestock and biofuel redirection (~ 7,800 kcal/day/person). Redirecting grain from cattle and biofuels to human consumption triples availability vs. 2a — the single most impactful intervention.
2c: Scenario 2c — + resilient food solutions (~ 10,000 kcal/day/person). Adding seaweed cultivation, crop relocation, industrial sugars, and SCP protein brings production to ~ 5× minimum requirements.
2d: Scenario 2d — + intensive greenhouse expansion and area increase (~ 14,000 kcal/day/person). Scaling greenhouses to 2.3M ha and cultivated area to 72M ha over 3 years. Surplus sufficient to support ~ 300M regional population.

Caveats
Key uncertainties

Results are sensitive to assumptions that require further research:

- Volcanic aerosol optical depth
- Regional agroclimate response
- Algae scale-up feasibility
- Nutritional adequacy of resilient diets
- International trade continuity
- Federal coordination heterogeneity
- Demographic nutritional variation
- Industrial system disruption depth

Yield loss estimates approximate for cold-sensitive crops (maize, soy) vs. cold-tolerant crops (potato, wheat, oat). Local edaphological and hydrological variation across Argentine provinces not yet captured.

Agenda
Future work

Priority research directions to reduce uncertainty and increase policy relevance:

- Argentina-specific agroclimate models calibrated to volcanic forcing scenarios
- Province-level food distribution and economic modeling
- Pilot trials for low-tech greenhouse deployment and algae cultivation in Patagonia
- Subgroup-specific nutritional needs analysis under ASRS conditions
- Comparative analysis: volcanic vs. nuclear ASRS forcing on Southern Cone food systems

At a glance
Key results

70% Population fed without adaptation (150 Tg)	3-6× Min. requirement met with full adaptation
300M People in the region who could receive surplus exports	7 yr Modeling window post-event

All scenarios modeled against a 2,100 kcal/day/person minimum requirement. Under the most severe scenario (150 Tg), full intervention brings net production to 7,800-14,000 kcal/day/person.

Case selection
Why Argentina?

Southern hemisphere location Temperate zone; less exposed to volcanic winter cooling concentrated in the Northern Hemisphere, as Tambora (1815) demonstrated.	Agricultural capacity 154M ha total agricultural land; 43M ha easily cultivable. Pre-ASRS gross production ~28,000 kcal/day/person.
Diversified economy Strong agri-industrial base; capacity to repurpose paper mills, biorefineries, and livestock infrastructure rapidly.	Regional connectivity Rail links with 6 neighbours; maritime trade with Brazil (key fossil fuel and food exporter), enabling surplus distribution.

Uruguay and Chile were also identified as priority candidates in the regional prioritization matrix. Argentina selected for its higher production capacity under modeled scenarios.

Discussion
Conclusions

- A volcanic ASRS of 150 Tg would reduce Argentina's net food production to ~ 70% of minimum population requirements - famine without intervention, even with rationing alone.
- Redirecting food from livestock and biofuels - the most impactful single measure - can alone increase human-available calories by up to 300%, bringing production above the survival threshold.
- A phased combination of all modeled interventions (2d) produces 3 – 6× minimum national requirements, enabling surplus export to support up to 300 million additional people in the region.
- Southern Cone countries occupy a structurally advantageous position in any ASRS scenario, including volcanic origin; advance national planning is both technically feasible and strategically urgent.
- Success depends on pre-event institutional preparedness: contingency plans, strategic input stockpiles, legislative flexibility, and inter-provincial coordination within Argentina's federal structure.

References & data
Selected references

[1] Xia et al. (2022). Global food insecurity and famine from reduced crop production due to nuclear war soot. *Nature Food*, 3, 586–596.
[2] Rivers et al. (2022). Food system adaptation and maintaining trade greatly mitigate global famine in ASRS. ALLFED Preprint.
[3] Alvarado et al. (2020). Scaling of greenhouse crop production in low sunlight scenarios. *Sci. Total Environ.*, 707.
[4] Wilson et al. (2023). Impact of the Tambora eruption of 1815 on islands. *Sci. Rep.*, 13, 3649.
[5] Ulloa Ruiz et al. (2024). Soluciones alimentarias resilientes... *REDER*, 8(2), 159–176.

Model code: <https://github.com/allfed/allfed-integrated-model>
Supplementary material: <https://doi.org/10.5281/zenodo.11658966>
Correspondence: juan@allfed.info